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Posteromedial Dynamic Balance Performance and Hip Muscle Activation in Healthy Young Women

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Objectives

Dynamic balance is an important component of physical function and mobility, particularly in populations at risk of functional decline and frailty.

We examined the association between normalized PM reach distance and lower-extremity muscle activation in healthy young women.

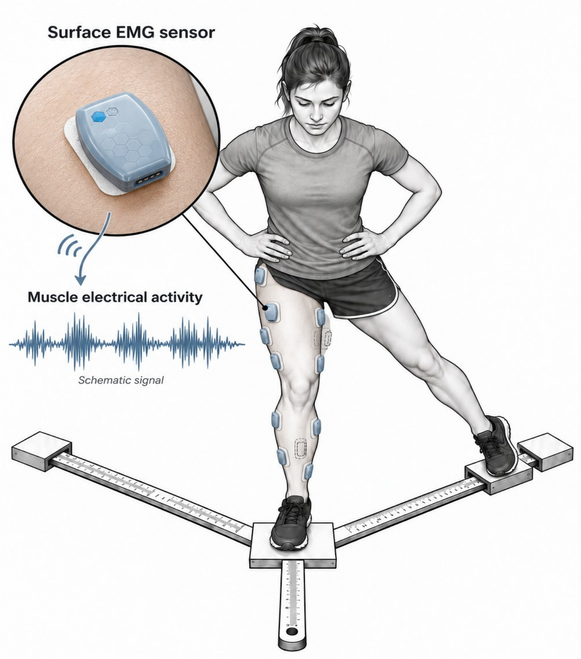
Methods

Participants: 50 healthy young women (22.9 ± 3.1 years). Normalized PM reach: 112.6 ± 14.1% (mean ± SD).

Y-Balance Test (YBT): Three trials per direction on the dominant limb. Maximum reach was normalized to leg length.

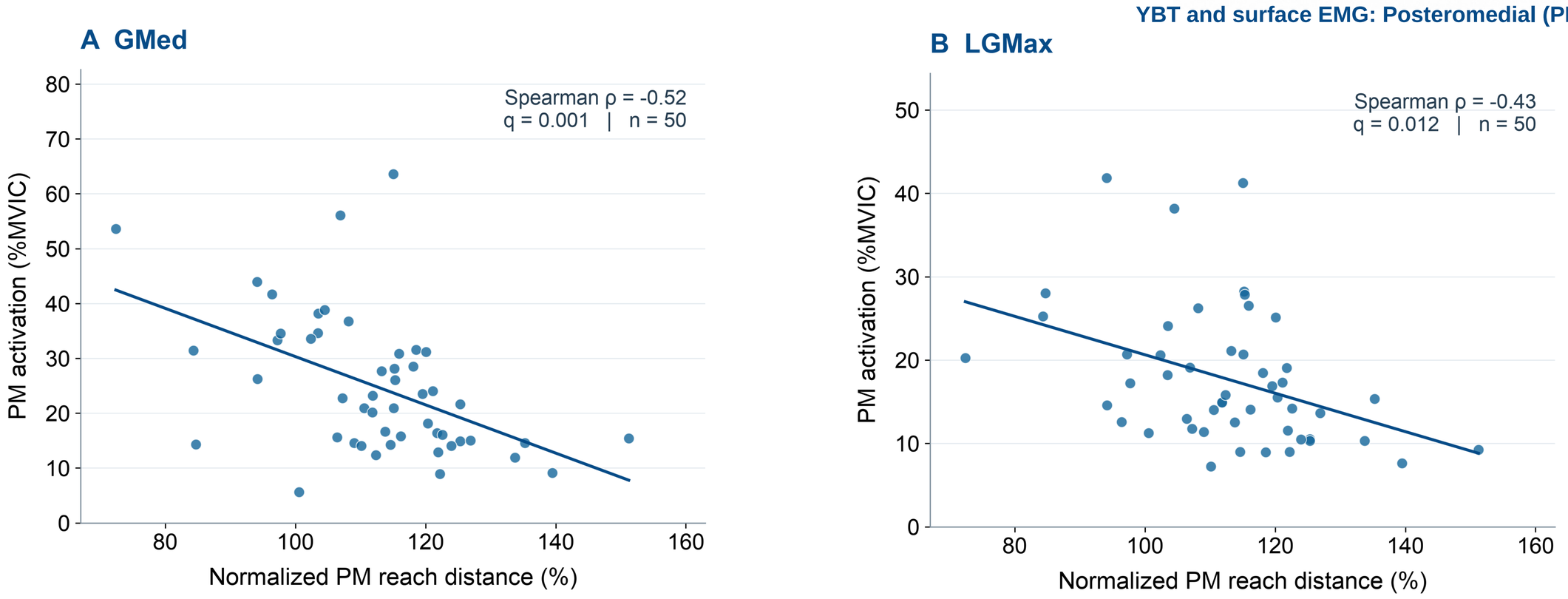
EMG: 12 muscles. Mean RMS from reach onset to maximum reach, averaged across three trials and expressed as %MVIC.

Analysis: Spearman correlations; FDR correction across 12 muscles for each activation measure.



YBT and surface EMG: Posteromedial (PM) reach

Results



PM activation relative to ANT and PL

Muscle	ρ	q
GMed	−0.41	.018
TFL	−0.45	.013

Relative activation = PM / mean(ANT, PL).
ANT: anterior; PL: posterolateral. Each point: one participant. Line: linear regression fit.
q: FDR-adjusted p value across 12 muscles per measure.
GMed: gluteus medius; LGMax: lower gluteus maximus;
TFL: tensor fasciae latae; MVIC: maximal voluntary isometric contraction.

Greater PM reach was associated with lower GMed and LGMax activation, and lower relative PM activation of GMed and TFL.

Conclusions

Greater PM reach performance was associated with lower activation of selected hip muscles, suggesting that better dynamic balance performance does not necessarily require greater muscle activation.

These findings may reflect differences in task-specific neuromuscular activation patterns among individuals with varying balance performance levels.

Interpretation and future research

The observed associations may be influenced by differences in muscle strength, joint mobility, and movement kinematics, which were not fully accounted for in the present analysis.

Future studies should investigate these factors and examine whether exercise-induced improvements in dynamic balance are accompanied by changes in muscle activation patterns.

Further research in older adults and clinical populations is warranted to determine the clinical relevance of these findings.

References

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2. Kaur N, et al. Lower extremity and trunk electromyographic muscle activity during performance of the Y-Balance Test on stable and unstable surfaces. Int J Sports Phys Ther. 2022;17:483–492.

Abstract No.: 0133 Conflicts of interest: None declared.

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